**CSA5108 Cryptography and Network security**

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**Exp 16 :** Monoalphabetic Substitution Cipher Frequency Attack
**Code:**

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

int main(void) {
   char cipher[5000], plain[5000];
   char english[]="ETAOINSHRDLCUMWFGYPBVKJXQZ";
   int freq[26]={0}, order[26], used[26]={0};
   char map[26];

   printf("Enter ciphertext: ");
   fgets(cipher,sizeof(cipher),stdin);

   for(int i=0;cipher[i];i++)
      if(isalpha((unsigned char)cipher[i]))
         freq[toupper((unsigned char)cipher[i])-'A']++;

   for(int i=0;i<26;i++) order[i]=i;

   for(int i=0;i<26;i++)
      for(int j=i+1;j<26;j++)
         if(freq[order[j]]>freq[order[i]]) {
            int t=order[i]; order[i]=order[j]; order[j]=t;
         }

   for(int i=0;i<26;i++)
      map[order[i]]=english[i];

   for(int i=0;cipher[i];i++) {
      if(isupper((unsigned char)cipher[i]))
```

```
            plain[i]=map[cipher[i]-'A'];
        else if(islower((unsigned char)cipher[i]))
            plain[i]=tolower((unsigned char)map[cipher[i]-'a']);
        else
            plain[i]=cipher[i];
    }

    plain[strlen(cipher)]='\0';
    printf("Frequency-based candidate:\n%s",plain);

    return 0;
}
```

